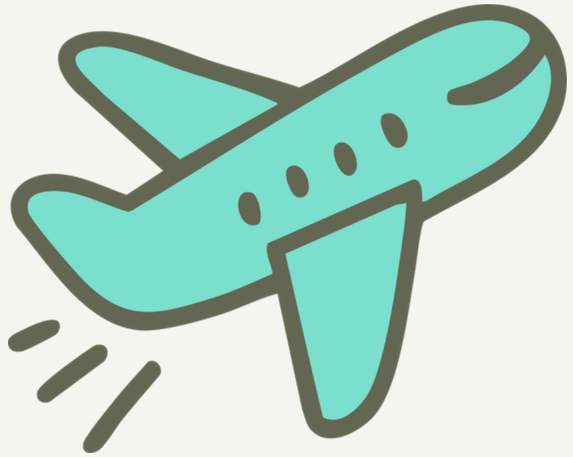
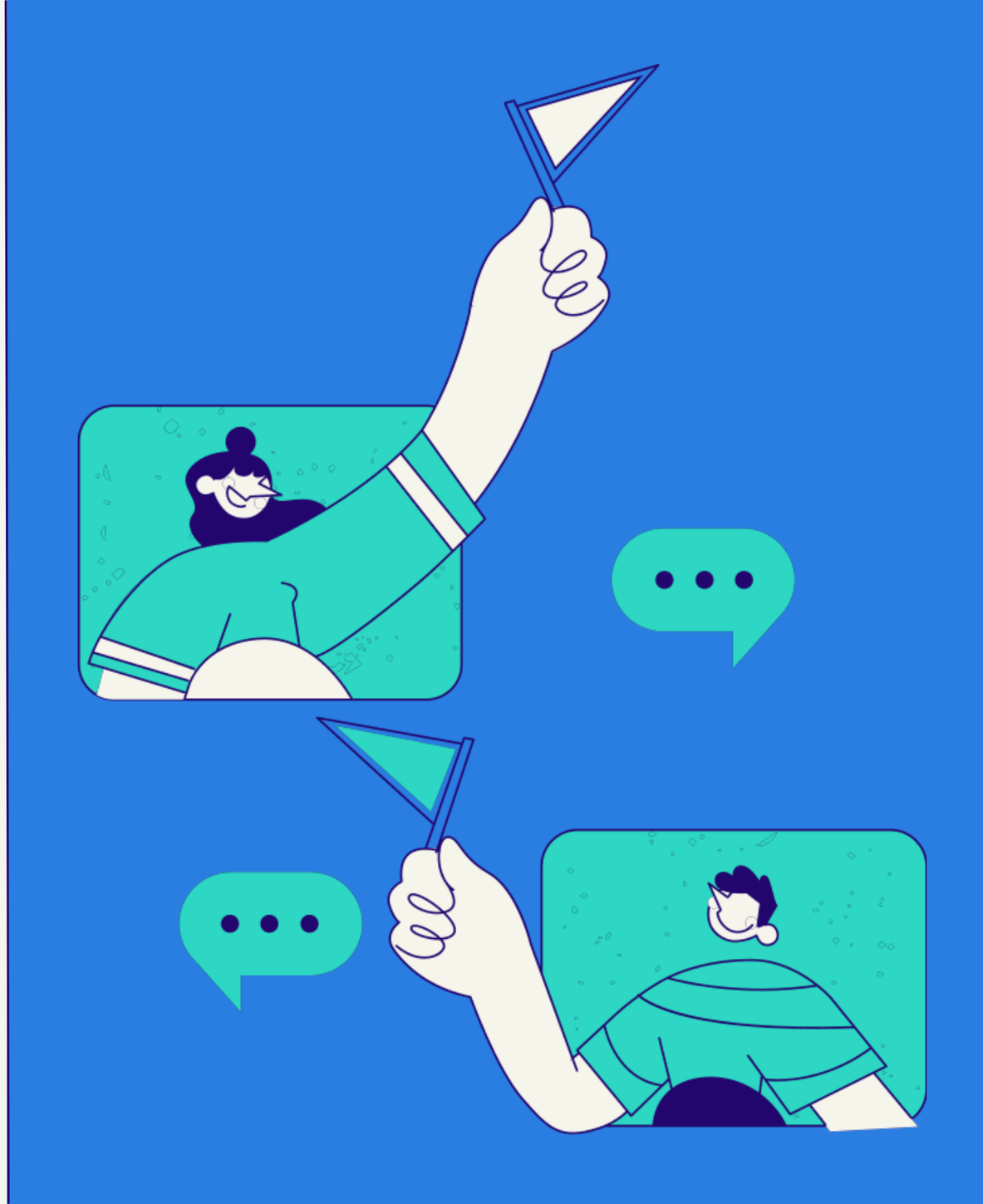
Safe Flight Booking System

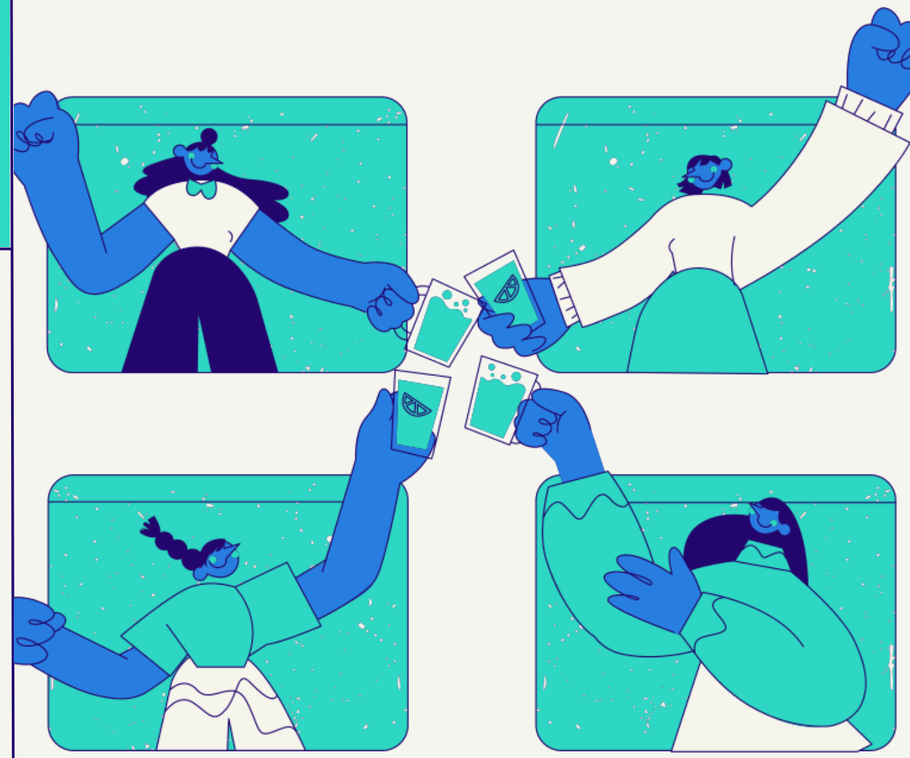
Presented by Group 5





PROJECT DESCRIPTION

- Develops the SafeFlight software with a focus on quality, verification, and validation
- Demonstrates application of Software Quality Assurance (SQA) across the full lifecycle
- Emphasizes process discipline, risk management, and quality control over just functionality
- Uses Agile Scrum methodology for development
- Integrates continuous testing to detect and prevent defects early



REQUIREMENTS



Requirements

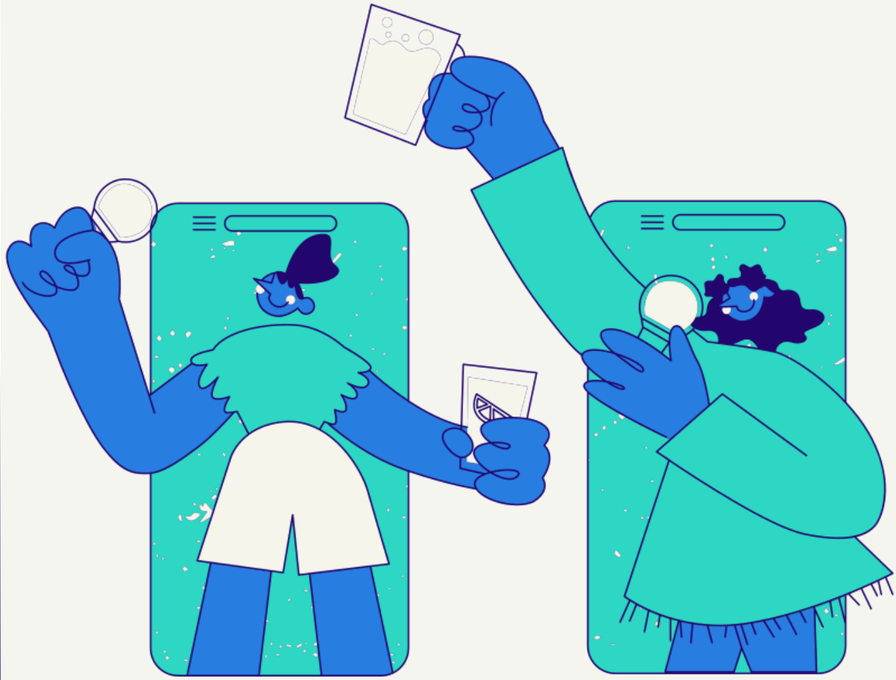
Requirement ID	Requirement Summary
R1	Email Validation: Handles mandatory presence, format(@), and domain checks
R2	Password Policy: Enforces 8-20 character limit and complexity
R3	Normalisation: Automatically trims whitespace and normalizes text case.
R4	Route Validation: Prevents Departure and Arrival being the same city.
R5	Temporal Constraints: Blocks past dates, System must accept Feb 29 only valid lead year and reject it otherwise.
R6	Passenger Name Rules: Restricts names to alphabetic characters only.
R7	Age Constraints: Validates age ranges (0-120) for safety and pricing.
R8	Contact Validation: Enforces exactly 10-digit numeric phone numbers.
R9	Baggage Calculation: Calculates fees for 0-50kgs; handles decimal/negative errors.
R10	Fare Discounts: Applies tiered % for Infants, Children, Seniors, and Defense.
R11	Seat Concurrency: Prevents race conditions(two users booking one seat.)
R12	Refund Policy: Tiered percentages based on-time-departure(3-day/1-day rules).

PROJECT MANAGMENT

- Methodology: Agile framework with defined Sprint backlog.

- Project Management: Utilizing Jira to break down high-level Epics into detailed User Stories and track task assignments.

- Version Control & Workflow: GitHub selected for version control, CI integration, and a firmly defined branching strategy.



CONFLUENCE DOCUMENTATION

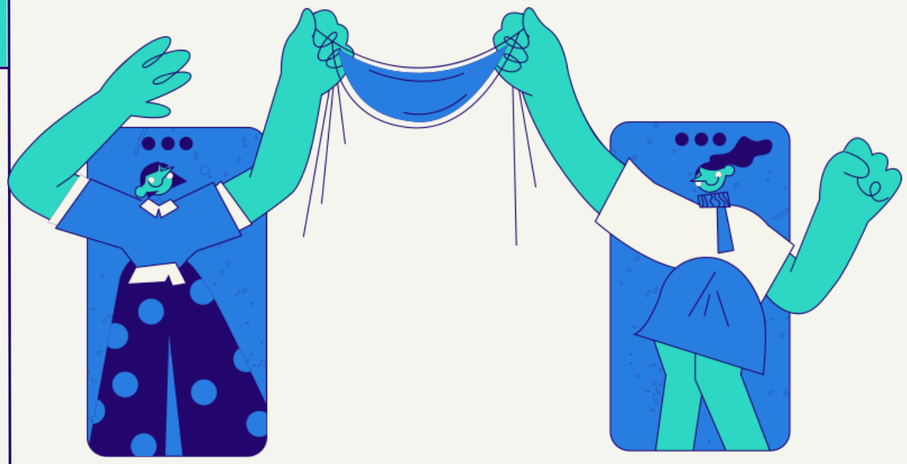


Confluence

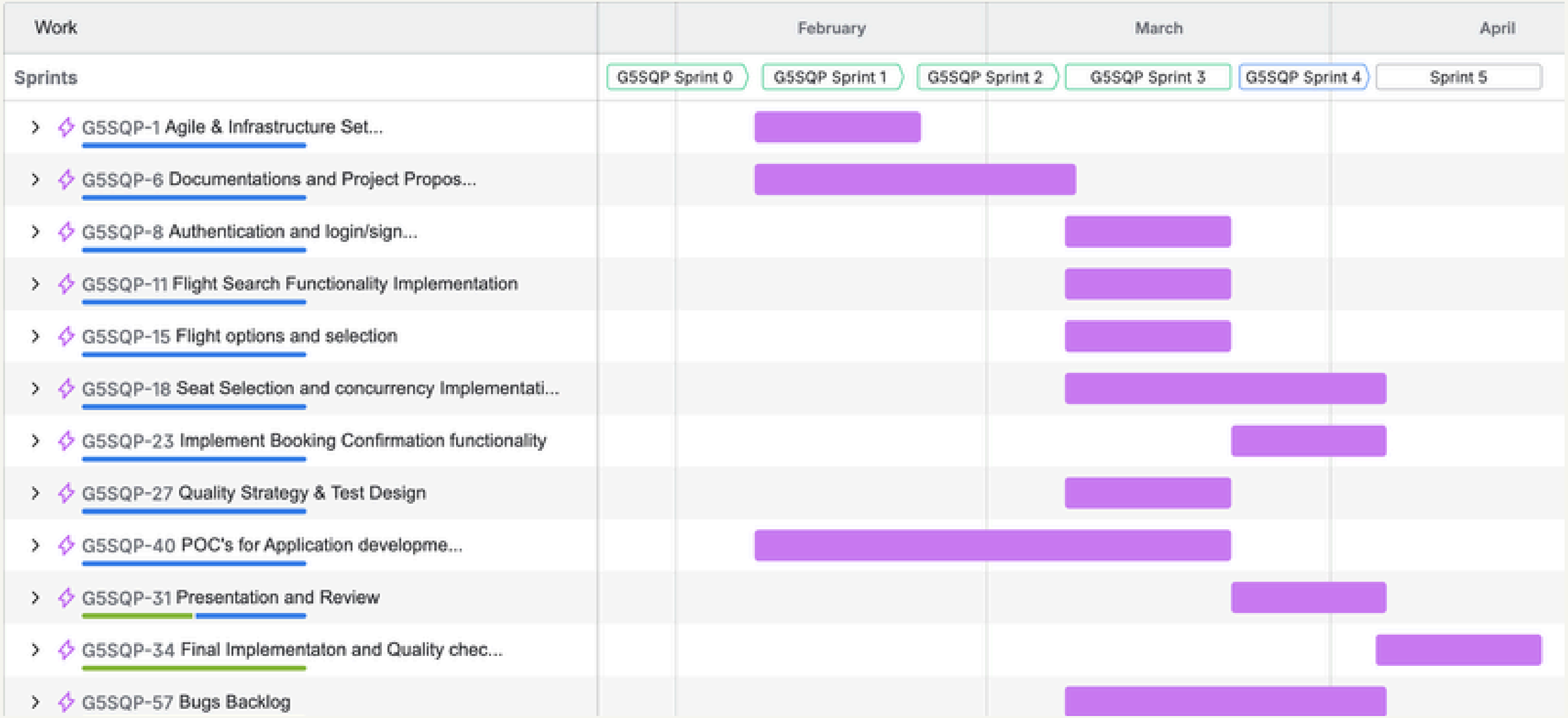
- DoR and DoD checklist
- Project Proposal
- Application Architecture D...
- Decision Tree- SafeFlight
- Minutes of Meerting
  - Week 1 (21-01-2026)
  - Week 2(28-01-2026)
  - Week 3(04-02-2026)
  - Week 4 (11-02-2026)
  - Week 5(18-02-2026)
  - Week 6 (25-02-2026)
  - Week 7 (04-03-2026)
  - Week 8 (12-03-2026)

SCRUM CEREMONIES

- The application was developed using Agile development practices, particularly Scrum.
- Development and other tasks were organized into short, time-boxed iterations called Sprints to facilitate incremental development and continuous verification.
- Conducted weekly stand-ups and the MOM was updates on Confluence



High Level Epics



Sprints(0-3)

G5SQP Sprint 0 25 Jan – 8 Feb (4 work items) — 0 | 12 | 0 — Complete sprint

G5SQP-41	POC for UI	POC'S FOR APPLICAT...	DONE	3
G5SQP-44	Integration of frontend with backend POC	POC'S FOR APPLICAT...	DONE	3
G5SQP-42	Backend POC	POC'S FOR APPLICAT...	DONE	3
G5SQP-43	Database POC	POC'S FOR APPLICAT...	DONE	3

+ Create

G5SQP Sprint 1 8 Feb – 22 Feb (5 work items) — 0 | 31 | 0 — Complete sprint

G5SQP-14	Validate Search input	FLIGHT SEARCH FUN...	DONE	5
G5SQP-10	Implement Login/logout functionality	AUTHENTICATION A...	DONE	8
G5SQP-39	Create System Architecture document	DOCUMENTATIONS A...	DONE	5
G5SQP-9	Creation of Sign up page	AUTHENTICATION A...	DONE	5
G5SQP-29	Identify high-risk modules	QUALITY STRATEGY ...	DONE	8

+ Create





G5SQP Sprint 2 22 Feb – 8 Mar (6 work items) — 0 | 39 | 0 — Complete sprint

G5SQP-13	Search flights functionality	FLIGHT SEARCH FUN...	DONE	8
G5SQP-12	Home page implementation for flight booking app	FLIGHT SEARCH FUN...	DONE	8
G5SQP-16	Display list of available flights	FLIGHT OPTIONS AN...	DONE	5
G5SQP-19	Select a flight option	FLIGHT OPTIONS AN...	DONE	5
G5SQP-28	Create Software Quality Assurance Plan	QUALITY STRATEGY ...	DONE	8
G5SQP-30	Document test cases	QUALITY STRATEGY ...	DONE	5

+ Create

G5SQP Sprint 3 8 Mar – 22 Mar (7 work items) — 0 | 24 | 0 — Complete sprint

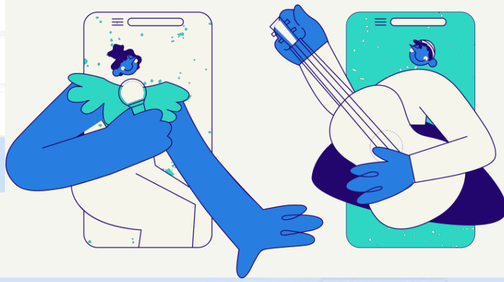
G5SQP-25	Impement Confirm booking and generate booking reference page	IMPLEMENT BOOKIN...	DONE	5
G5SQP-21	Prevent double booking	SEAT SELECTION AN...	DONE	3
G5SQP-24	Implement review booking summary page	IMPLEMENT BOOKIN...	DONE	5
G5SQP-26	View confirmation page (post-booking)	IMPLEMENT BOOKIN...	DONE	5
G5SQP-64	Styling and html pages for search flight page		DONE	3
G5SQP-66	Styling and html for my bookings page		DONE	3
G5SQP-59	search results page showing error even when flights are listed for the day	BUGS BACKLOG	DONE	-

Sprints(4&5)

G5SQP Sprint 4 23 Mar – 5 Apr (13 work items) — 0 | 26 | 5 — Complete sprint

G5SQP-60	User is able to select flights from past days	BUGS BACKLOG	DONE	-	KS
G5SQP-61	Inconsistant mapping for discounts for different categories	BUGS BACKLOG	DONE	-	KS
G5SQP-63	Styling for destop view for application signup and login page		DONE	5	Y
G5SQP-22	Validate luggage/passenger constraints	SEAT SELECTION AN...	DONE	3	KS
G5SQP-70	Styling and html for signup page.		DONE	3	Y
G5SQP-32	Build presentation deck	PRESENTATION AND ...	IN QA	3	AS
~~G5SQP-55~~	Demo and code review evidence doc	PRESENTATION AND ...	NOT STARTED	5	SA
G5SQP-62	User is not able to book ticket without selecting any extra luggage option	BUGS BACKLOG	READY FOR DONE	-	HK
G5SQP-71	When searching for flights the city names are case sensitive	BUGS BACKLOG	READY FOR DONE	-	HK
G5SQP-68	Styling and html for search and search results page		DONE	3	Y
G5SQP-69	Styling and html for login page.		DONE	3	Y
G5SQP-65	Styling and html for confirm booking page		DONE	3	MU
G5SQP-67	Styling and html for cancel booking and other pages		DONE	3	MU

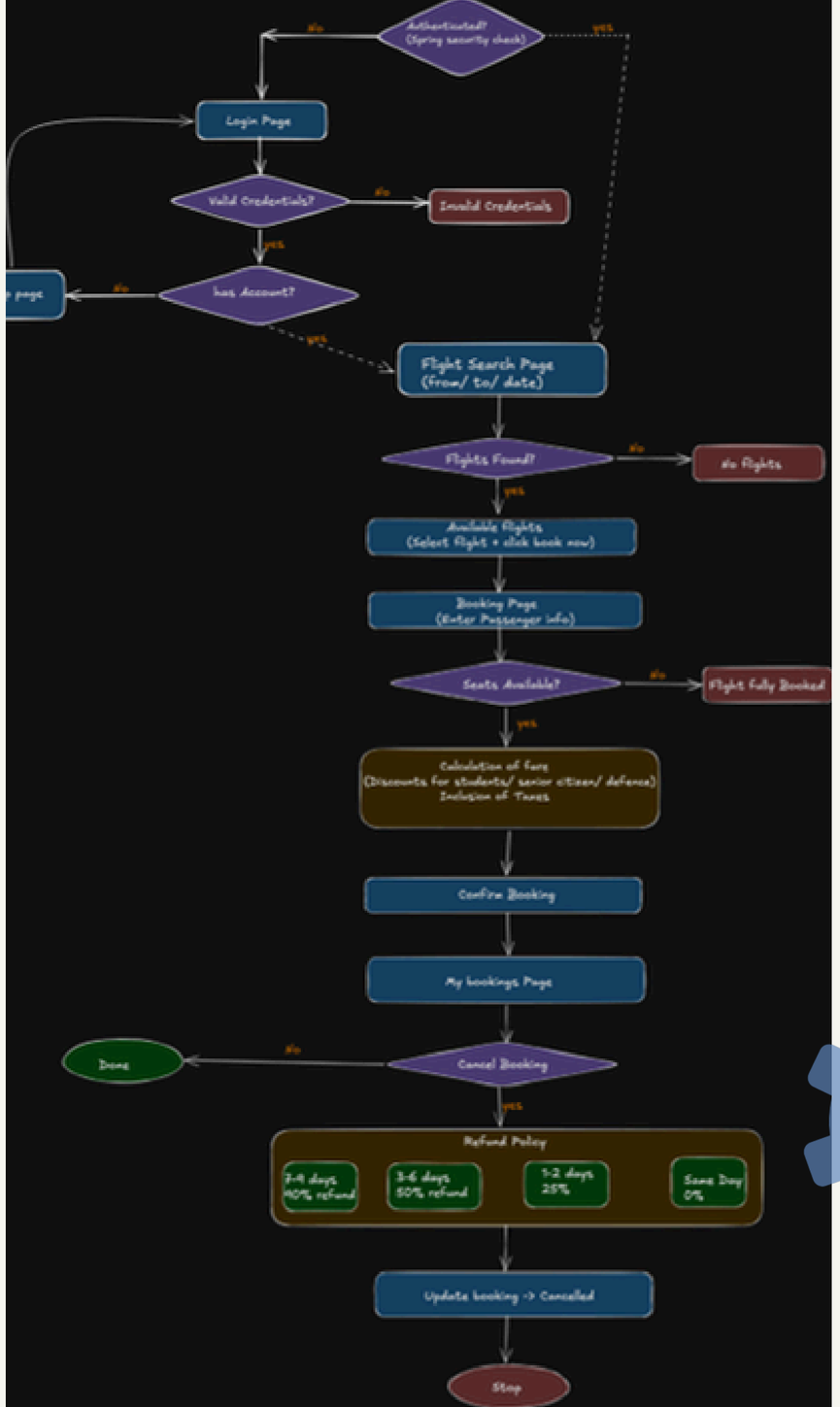
+ Create



Sprint 5 5 Apr – 19 Apr (4 work items) — 0 | 0 | 26 — Start sprint

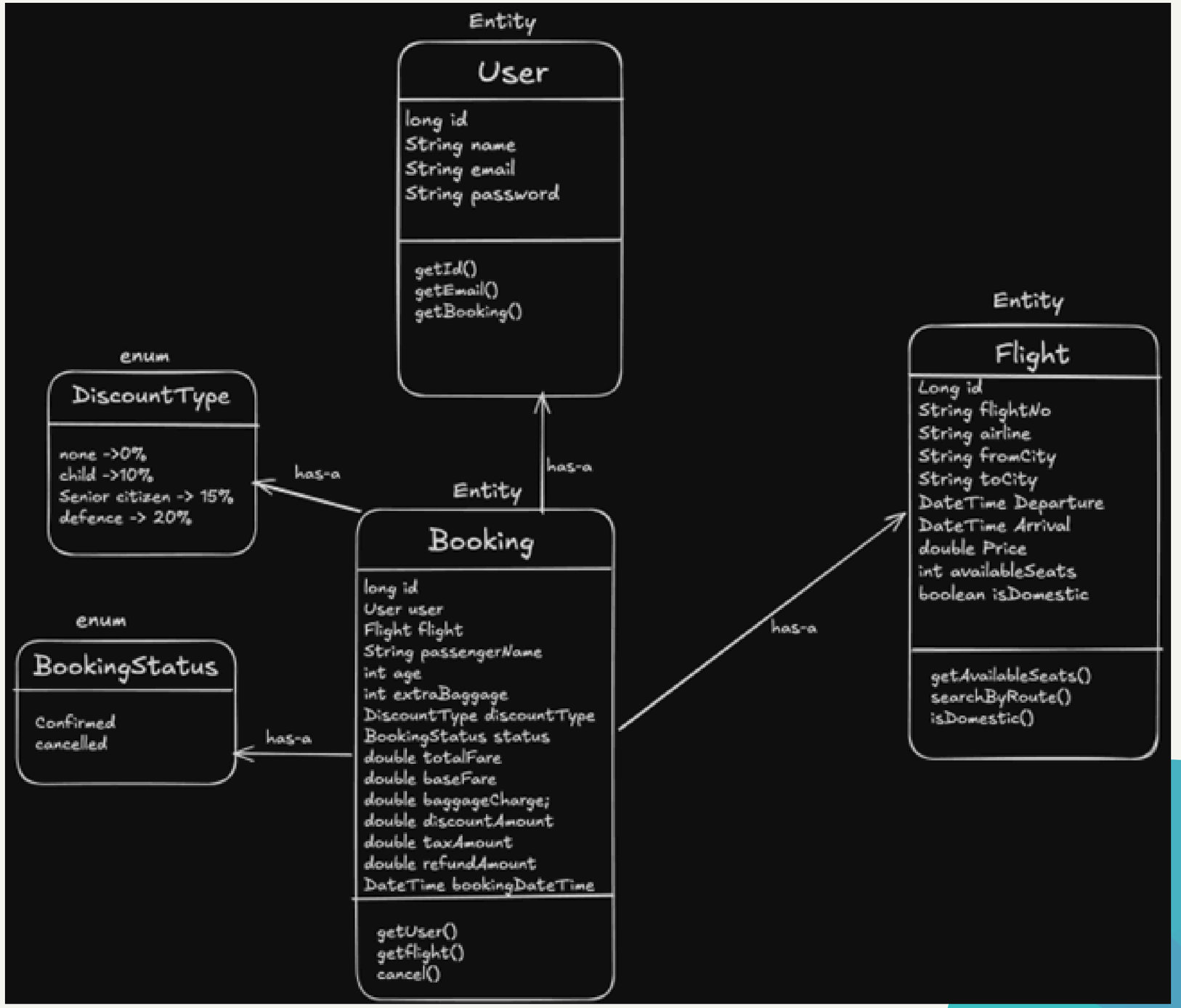
~~G5SQP-35~~	Complete all User stories	FINAL IMPLEMENTAT...	NOT STARTED	5
~~G5SQP-36~~	Complete Unit tests coverage	FINAL IMPLEMENTAT...	NOT STARTED	8
~~G5SQP-37~~	Static analysis using Sonarqube	FINAL IMPLEMENTAT...	NOT STARTED	8
~~G5SQP-38~~	Security testing	FINAL IMPLEMENTAT...	NOT STARTED	5

+ Create

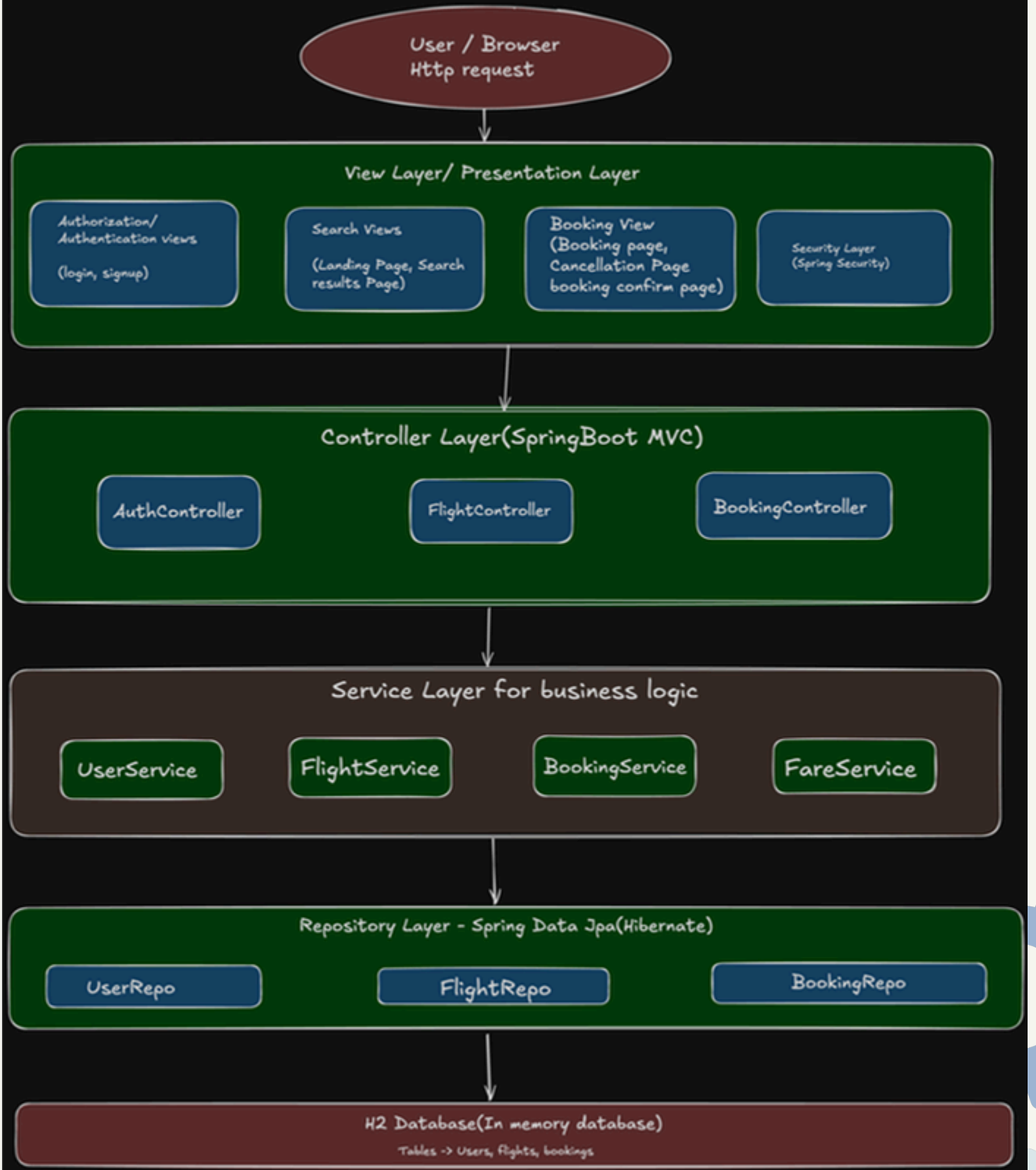


System Flow diagram

LINK



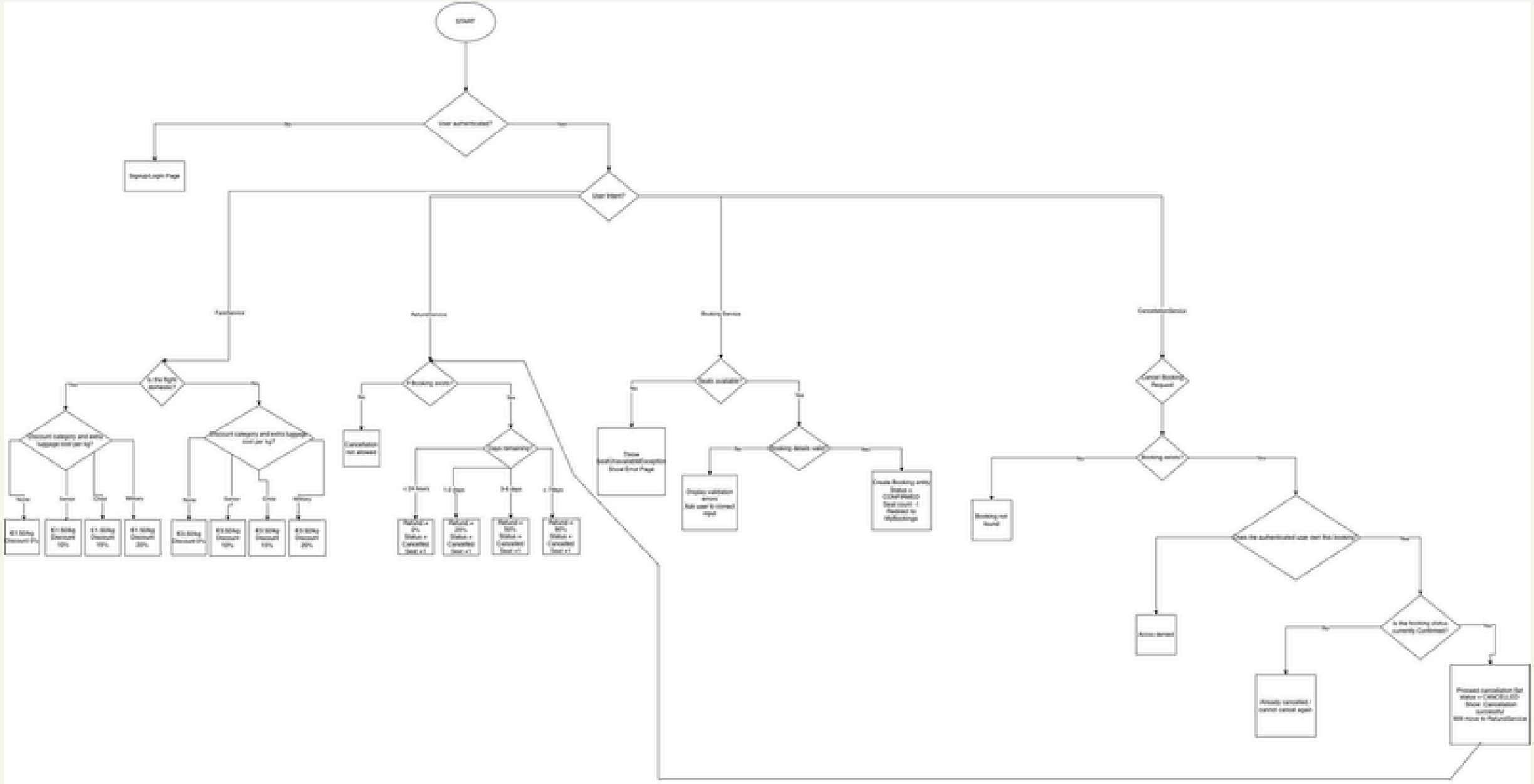
Class diagram



MVC architecture

LINK

Decision Diagram



TEST STRATAGIES

The Software Quality Assurance Plan (SQAP) establishes the testing strategy, standards, tools, roles, and deliverables that govern verification and validation activities for the Safeflight booking system. It ensures all the software components meet defined functional and non-functional requirements.

This plan applies to all modules developed under the Safeflight project including:
* Backend controls: AuthController, FlightController, BookingController
* Business Logic: UserService, FlightService, BookingService, FareService.
* Data Access Layer: UserRepository, FlightRepository, BookingRepository
* Web Layer: Thymeleaf templates, Spring Security Configuration.
* Datamodels and DTOs: User, Flight, Booking, DiscountType, BookingStatus.

Level	Scope	Ownership	Primary Goal
Unit Testing	Individual class/ method	Developer	Verify each component in isolation with mocked dependencies
Integration Testing	Service+Repository +DB	Developer/QA	Verify that components interact correctly when wired together
System Testing	End-to-end application	QA Engineer	Validate complete user workflows against acceptance criteria

Requirements

Requirement ID	Requirement Summary
R1	Email Validation: Handles mandatory presence, format(@), and domain checks
R2	Password Policy: Enforces 8-20 character limit and complexity
R3	Normalisation: Automatically trims whitespace and normalizes text case.
R4	Route Validation: Prevents Departure and Arrival being the same city.
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R10	Fare Discounts: Applies tiered % for Infants, Children, Seniors, and Defense.
R11	Seat Concurrency: Prevents race conditions(two users booking one seat.)
R12	Refund Policy: Tiered percentages based on-time-departure(3-day/1-day rules).

[Test plan Document]

Test cases

Summary | Search | Sort key | Search and add test cases

11 records

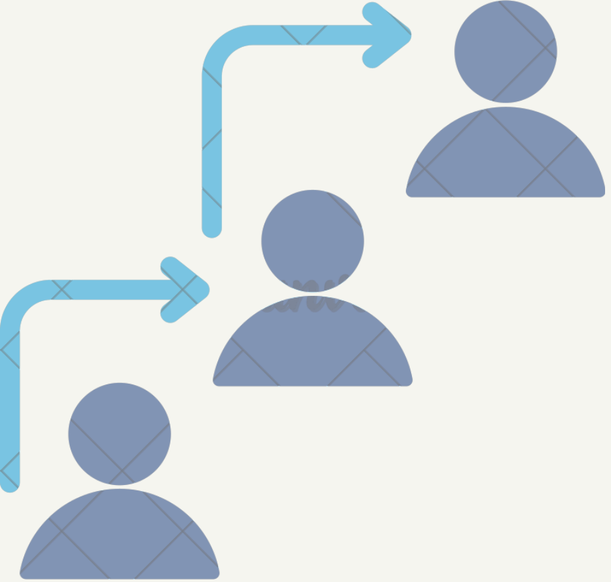
TC-(99337: Test Case (BB-BK-01-E2E Domestic Booking Calculation (Good Scenario)) CONFIRMED high Fare Discount(R10) shresha.lakshmanJ shresha.lakshmanJ

TC-(99338: Test Case (BB-BK-02: Concurrency - Race Condition for Last Seat (Best Scenario)) CONFIRMED high Seat Concurrency(R11) shresha.lakshmanJ shresha.lakshmanJ

TC-(99339: Test Case (BB-BK-03: Security - Unauthenticated Access Attempt (Worst Scenario)) CONFIRMED high Authentication and Access(R1,R2,R3) shresha.lakshmanJ shresha.lakshmanJ

TC-(99340: Test Case (BB-AU-01 Security - SQL Injection in Signup (Worst Scenario)) CONFIRMED high Authentication and Access(R1,R2,R3) shresha.lakshmanJ shresha.lakshmanJ

TC-(99341: Test Case (BB-CN-01 Unauthorized Cancellation Attempt (Worst Scenario)) CONFIRMED high Authentication and Access(R1,R2,R3) shresha.lakshmanJ shresha.lakshmanJ



CI PIPLINE

- Build runs at every commit
- Unit test runs at every commit
- Static code analysis using sonarcloud

Code | Blame | 29 lines (24 loc) · 631 Bytes

```yaml
name: CI Pipeline

on:
  push:
    branches: [ "master", "develop" ]
  pull_request:
    branches: [ "master" ]

jobs:
  build-test-sonar:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4
        with:
          fetch-depth: 0

      - name: Set up JDK 17
        uses: actions/setup-java@v4
        with:
          distribution: 'temurin'
          java-version: '17'

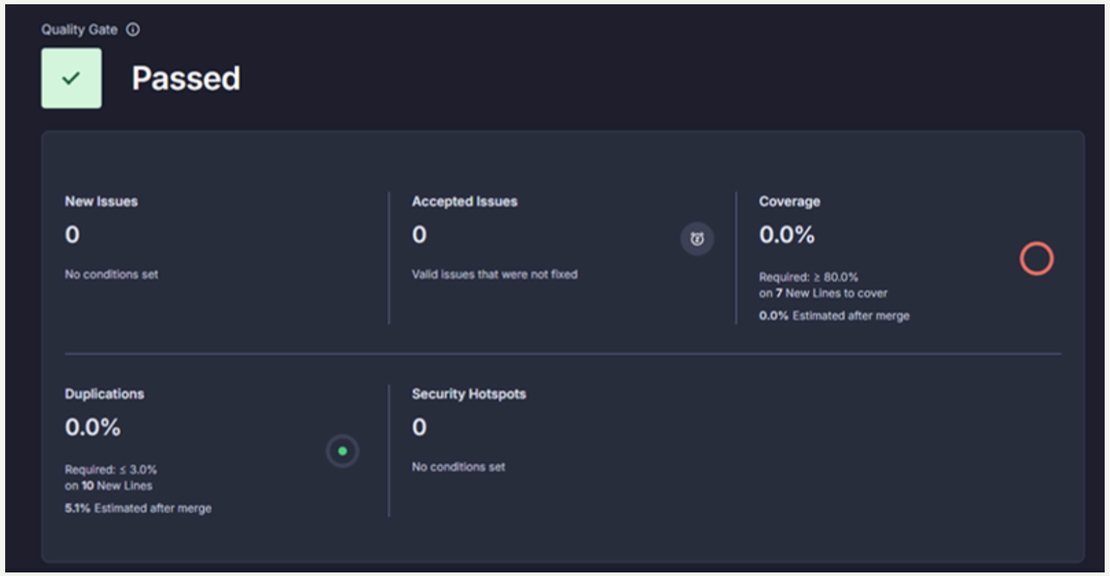
      - name: Build, test, and analyze
        working-directory: backend
        env:
          SONAR_TOKEN: ${{ secrets.SONAR_TOKEN }}
        run: mvn clean verify sonar:sonar -Dsonar.qualitygate.wait=false
```

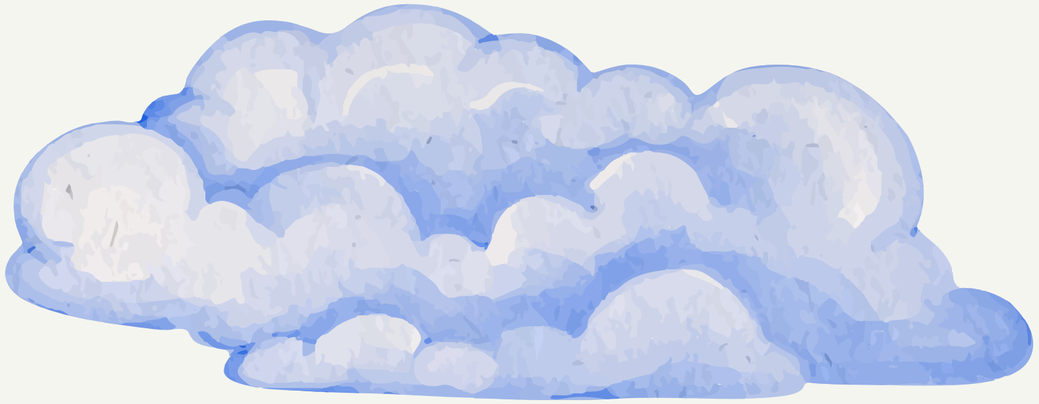
build-test-sonar

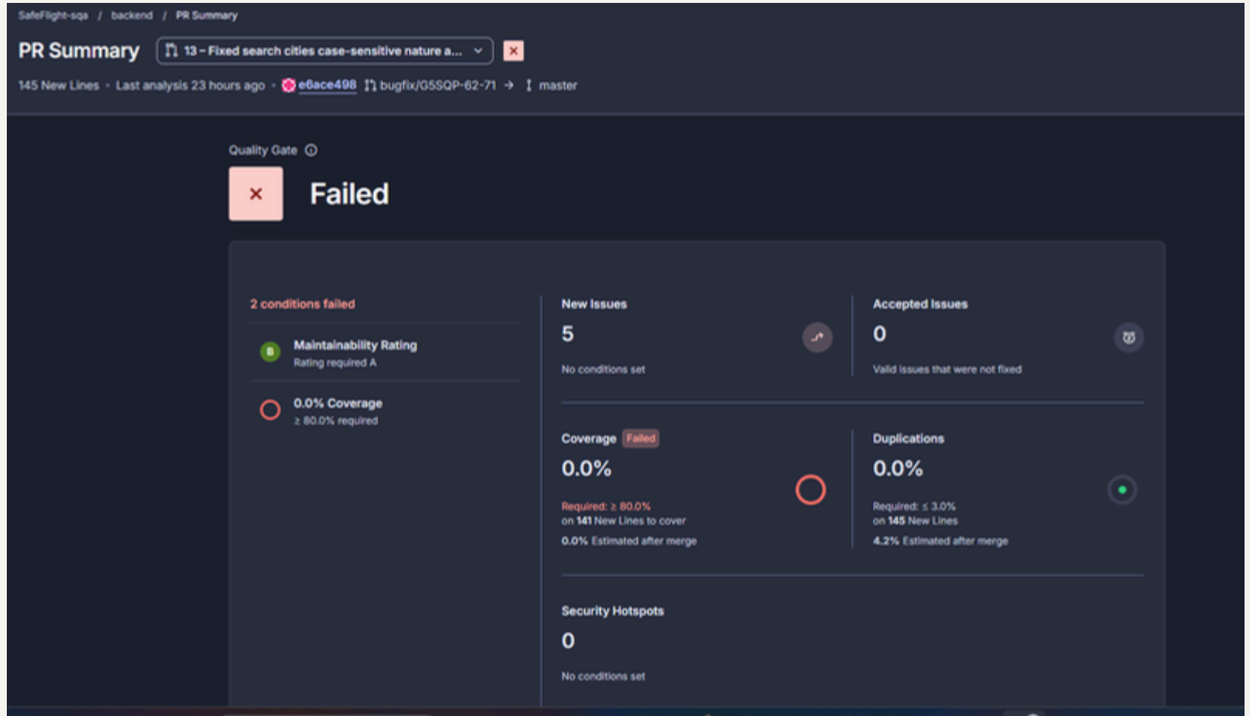
succeeded 14 hours ago in 1m 13s

- Set up job
- Checkout code
- Set up JDK 17
- Build, test, and analyze
- Post Set up JDK 17
- Post Checkout code
- Complete job

STATIC ANALYSIS - SonarCloud

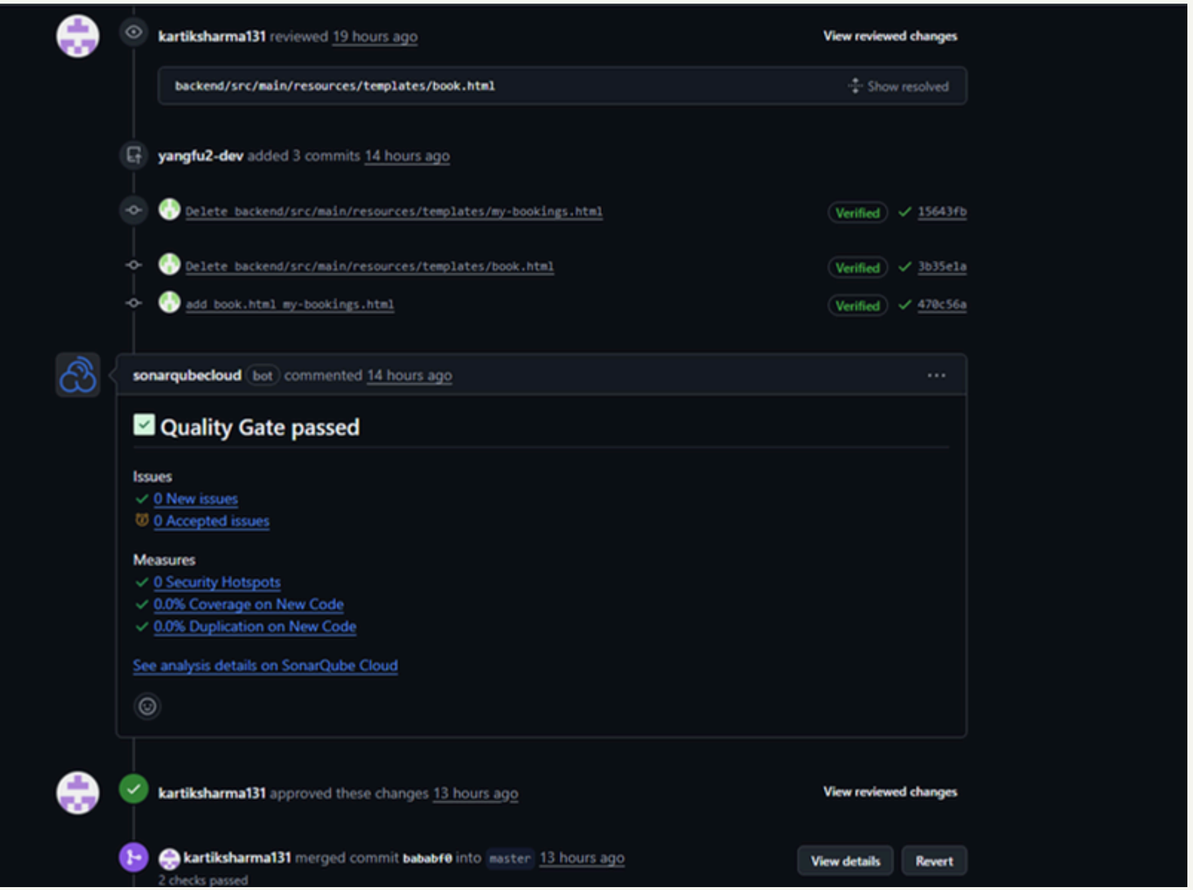


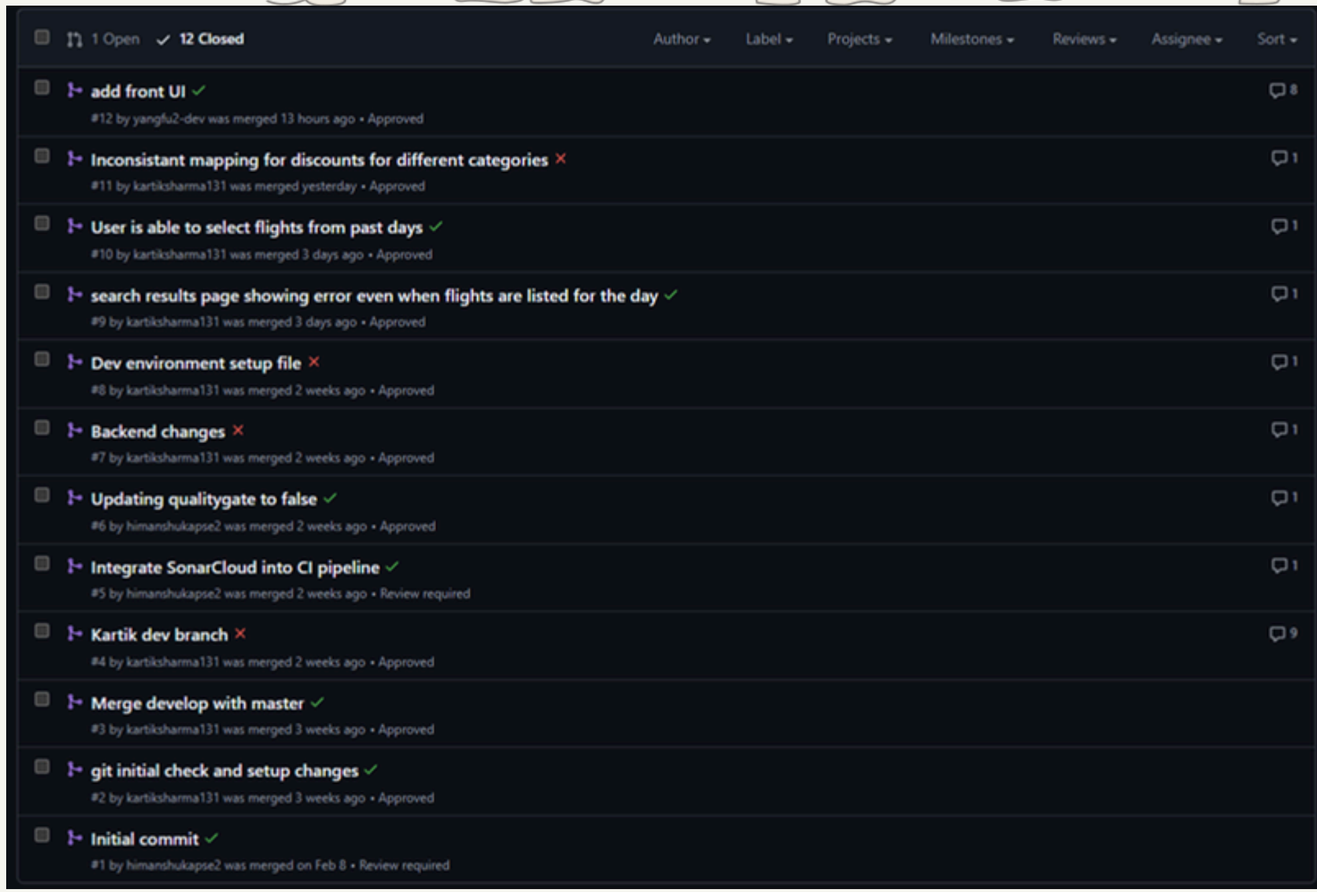




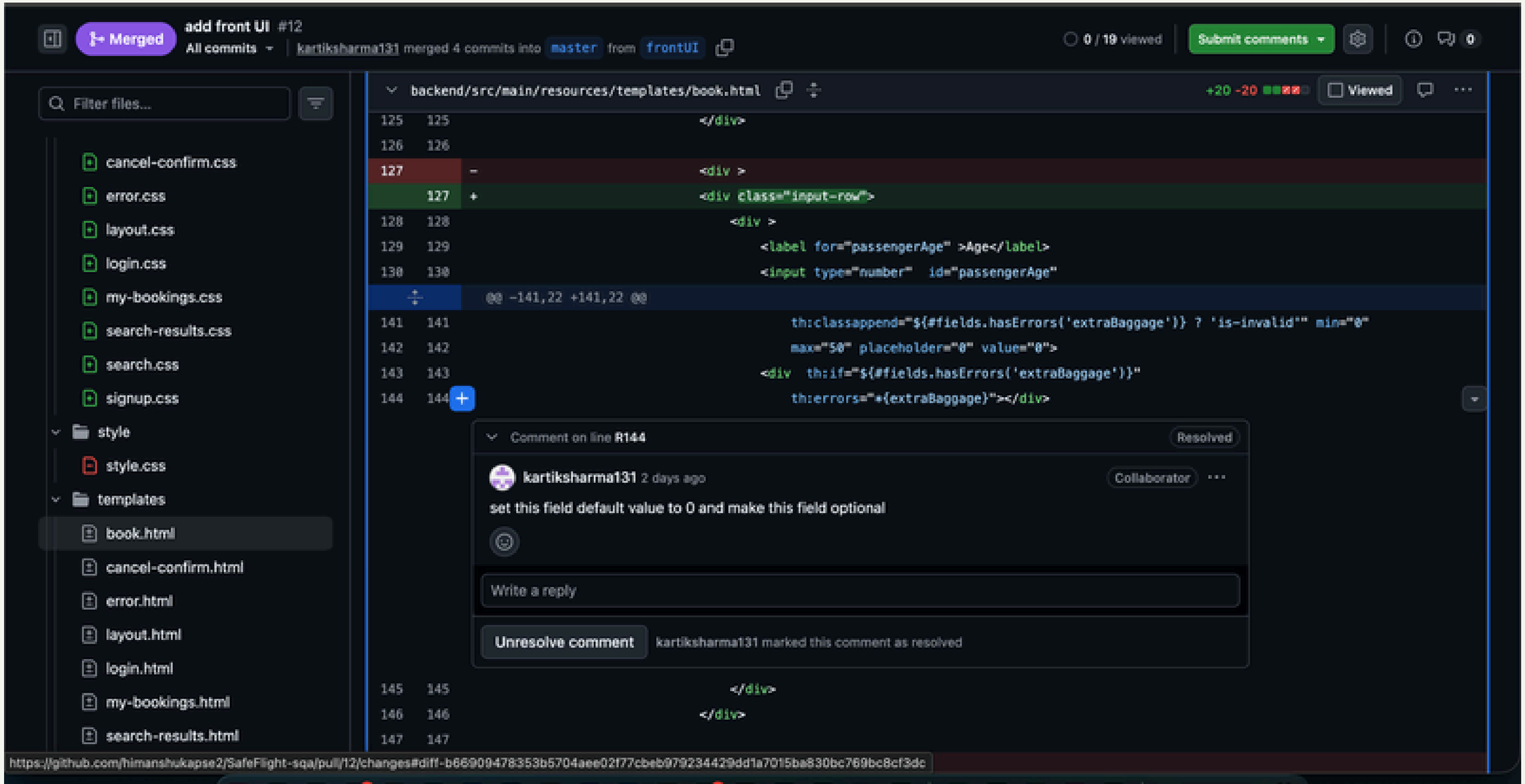
Pull Requests & Code Review







Pull Requests & Code Review



Bugs discovered in development phase

Spaces / Group-5 Software Qua... / G5SQP-57

1

Bugs Backlog

Description

Edit description

Child work items

0% Done

Work	Priority	Assi...	Status
G5SQP-60 User is able to select flights from past days	M...	K..	DONE
G5SQP-61 Inconsistant mapping for discounts for different categories	M...	K..	DONE
G5SQP-59 search results page showing error even when flights are listed for the day	M...	K..	DONE
G5SQP-62 User is not able to book ticket without selecting any extra luggage option	M...	H.	READY FOR DONE
G5SQP-71 When searching for flights the city names are case sensitive	M...	H.	READY FOR DONE



Tasks for upcoming Sprints...

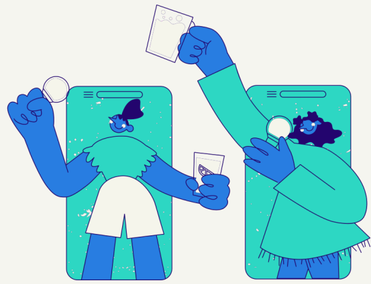
- Booking return flights functionality at the time of booking.
- Integrate Mend for vulnerability analysis in CI.
- Unit Test coverage >=80%.
- Seat selection pricing strategy.
- Regression test suite.
- Automation implementation.





Team Members, and Roles.

Project Management & Scrum Master	Kartik S
Backend	Kartik S, Himanshu K
Database	Kartik S, Himanshu K
Frontend	Yang F, Mohammad U
Integration	Kartik S, Angad S, Himanshu K
Deployment & GitHub Workflow	Himanshu K, Angad S
Testing	Shresha L, Himanshu K
Documentation	Angad S



Appeasing the Demo Gods

OutOfMemoryError:

End of Presentation
Thank you!




Q & A